

Lesson 13: Hypothesis Testing 1: Single-sample mean

TB sections 4.3, 5.1

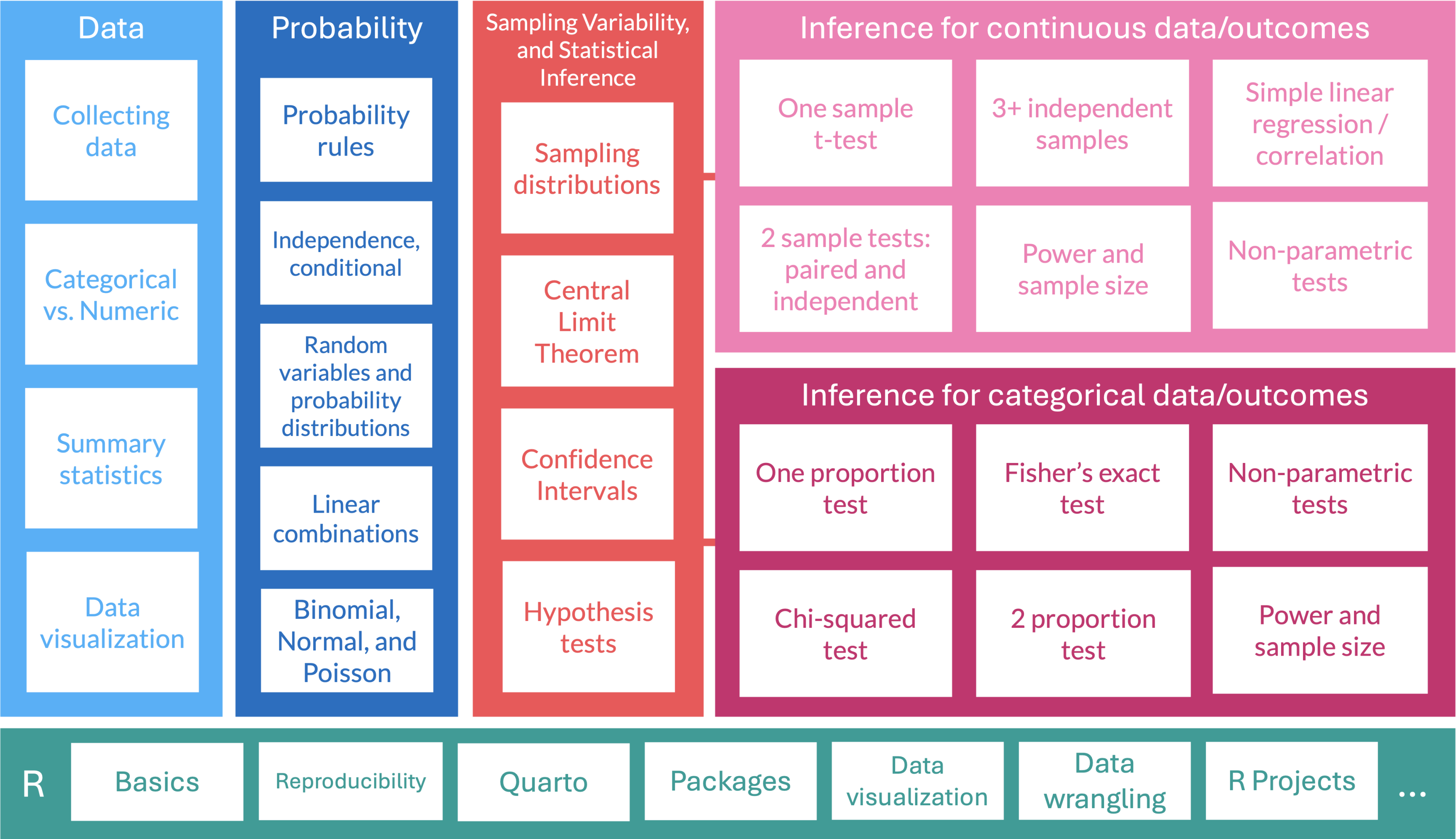
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Learning Objectives

1. Understand the relationship between point estimates, confidence intervals, and hypothesis tests.
2. Determine if a single-sample mean is different than a population mean using a hypothesis test.
3. Use R to calculate the test statistic, p-value, and confidence interval for a single-sample mean.

Where are we?



Hypothesis tests we will learn

CI's and hypothesis testing for different scenarios:

Lesson	Section	Population parameter	Symbol (pop)	Point estimate	Symbol (sample)
12	5.1	Pop mean	μ	Sample mean	$\bar{x}$
13	5.2	Pop mean of paired diff	μ_d or δ	Sample mean of paired diff	$\bar{x}_d$
14	5.3	Diff in pop means	$\mu_1 - \mu_2$	Diff in sample means	$\bar{x}_1 - \bar{x}_2$
16	8.1	Pop proportion	p	Sample prop	$\hat{p}$
16	8.2	Diff in pop prop's	$p_1 - p_2$	Diff in sample prop's	$\hat{p}_1 - \hat{p}_2$

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Answering a research question with a single mean

Research question is a generic form: Is there evidence to support that the population mean is different than μ_0 ?

- μ_0 is the generic form for a prescribed value: μ is the population mean and we want to see if $\mu = \mu_0$

Two approaches to answer this question:

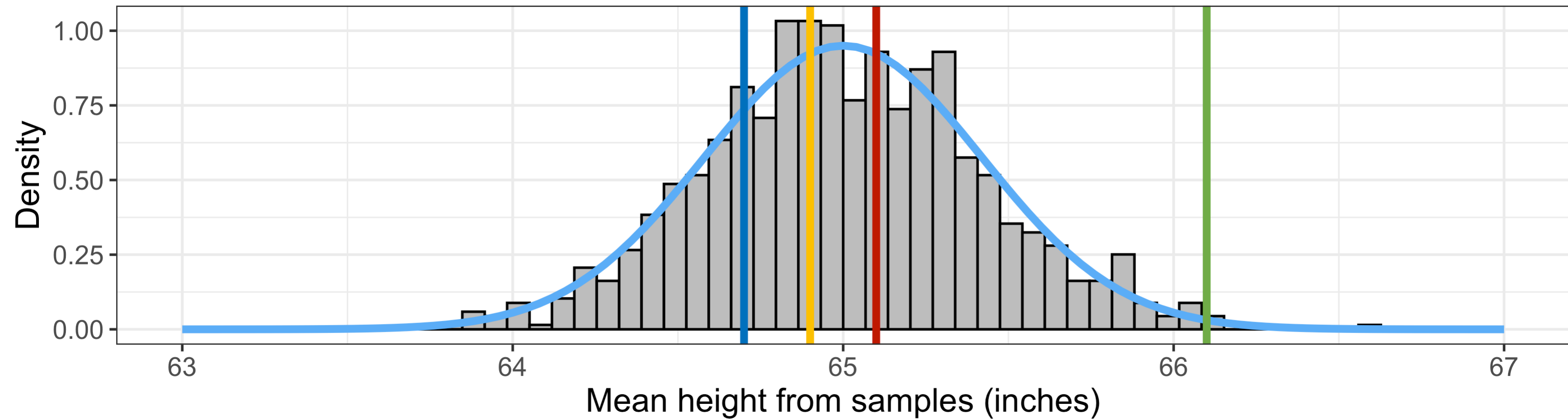
Confidence interval

- Create a **confidence interval (CI)** for the population mean μ from our sample data and determine whether a prescribed value (μ_0) is inside the CI or not
- Answering the question: is μ_0 a plausible value given our data?
 - If CI does NOT contain μ_0 then the answer is “Yes, evidence to support population mean is different than μ_0 ”

Hypothesis test

- Run a **hypothesis test** to see if there is evidence that the population mean μ is *significantly different* from a prescribed value (μ_0)
- This does not give us a range of plausible values for the population mean μ .
- Instead, we calculate a *test statistic* and *p-value*
- See how likely we are to observe the sample mean $\bar{x}$ or a more extreme sample mean assuming that the population mean μ is a prescribed value

Last last time: Point estimates



Sample 50 people
 $\bar{x} = 65.1, s = 2.8$

Sample 50 people
 $\bar{x} = 64.7, s = 3.1$

Sample 50 people
 $\bar{x} = 64.9, s = 3.2$

Sample 50 people
 $\bar{x} = 66.1, s = 3.4$

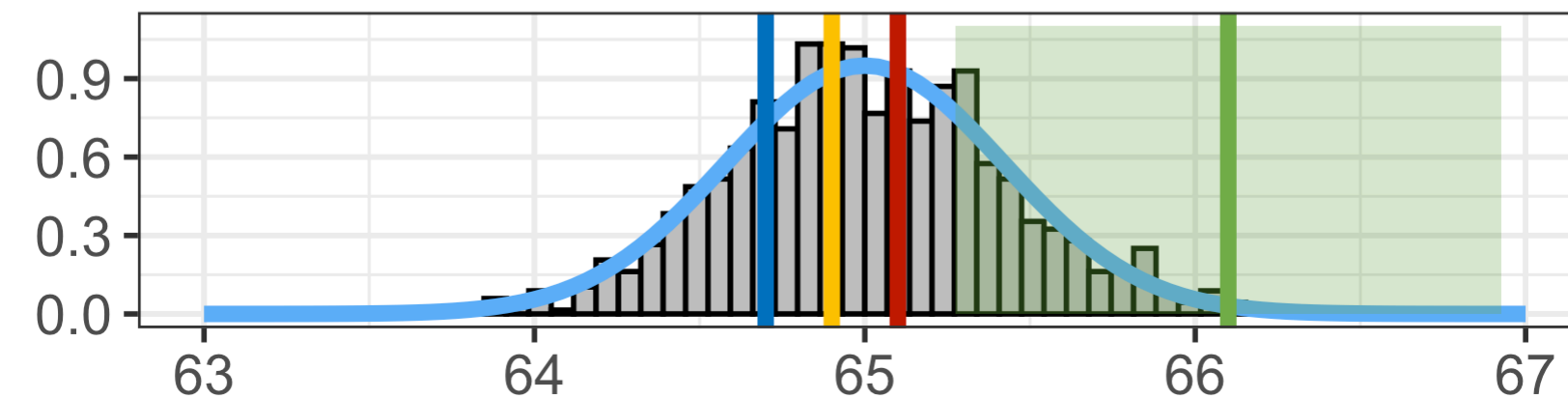
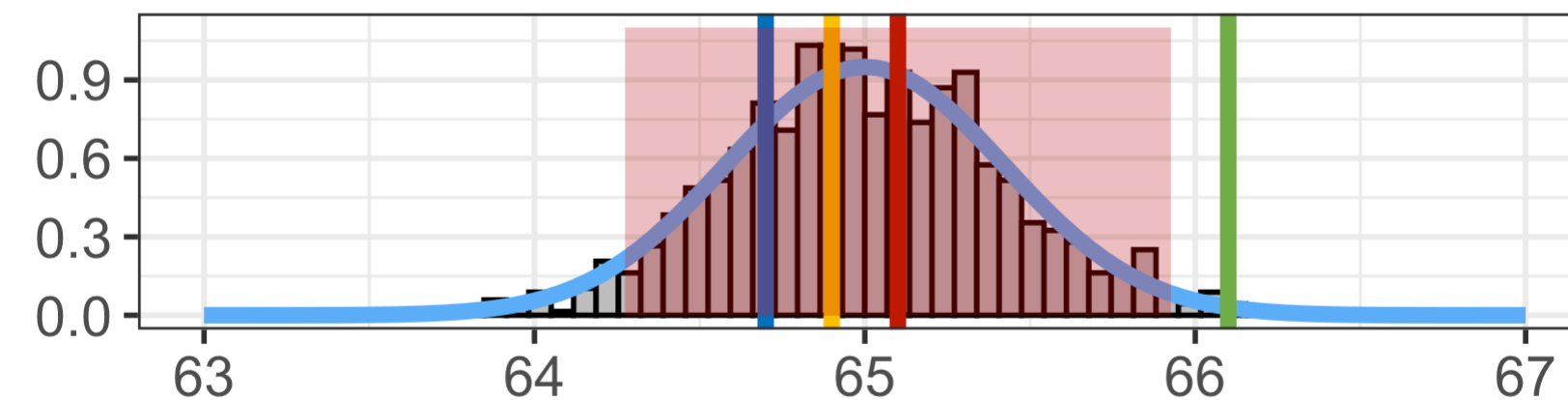
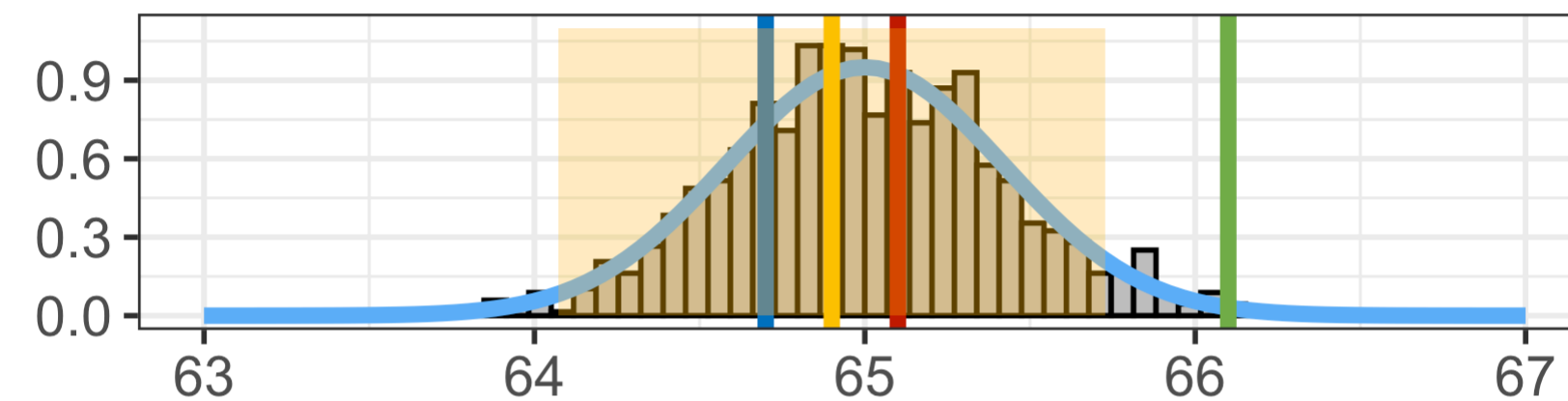
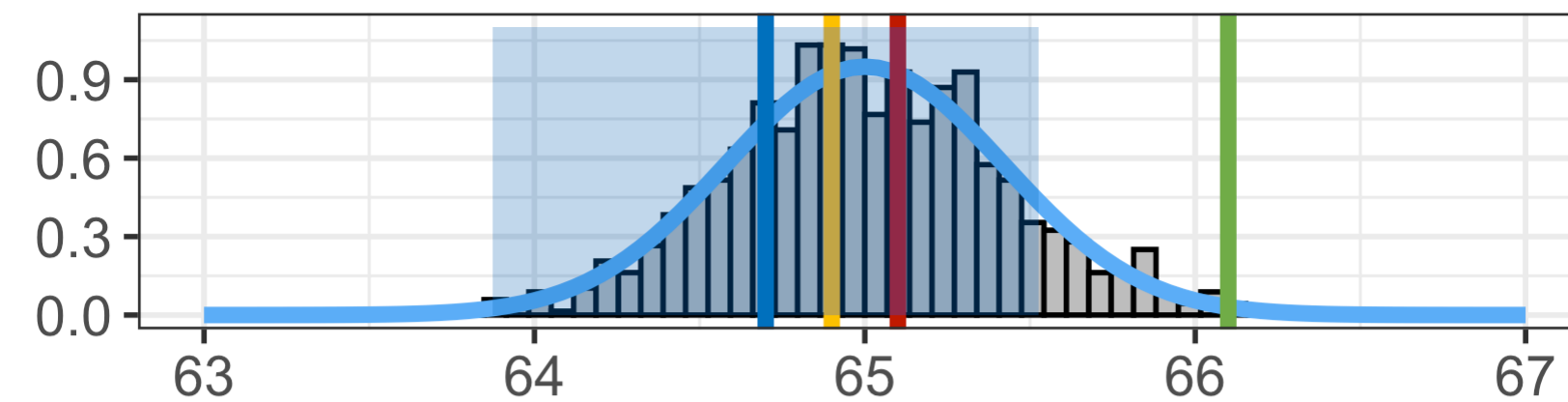
Last time: Point estimates with their confidence intervals for μ

Sample 50 people
 $\bar{x} = 64.7, s = 3.1$

Sample 50 people
 $\bar{x} = 64.9, s = 3.2$

Sample 50 people
 $\bar{x} = 65.1, s = 2.8$

Sample 50 people
 $\bar{x} = 66.1, s = 3.4$



Do these confidence intervals include μ ?

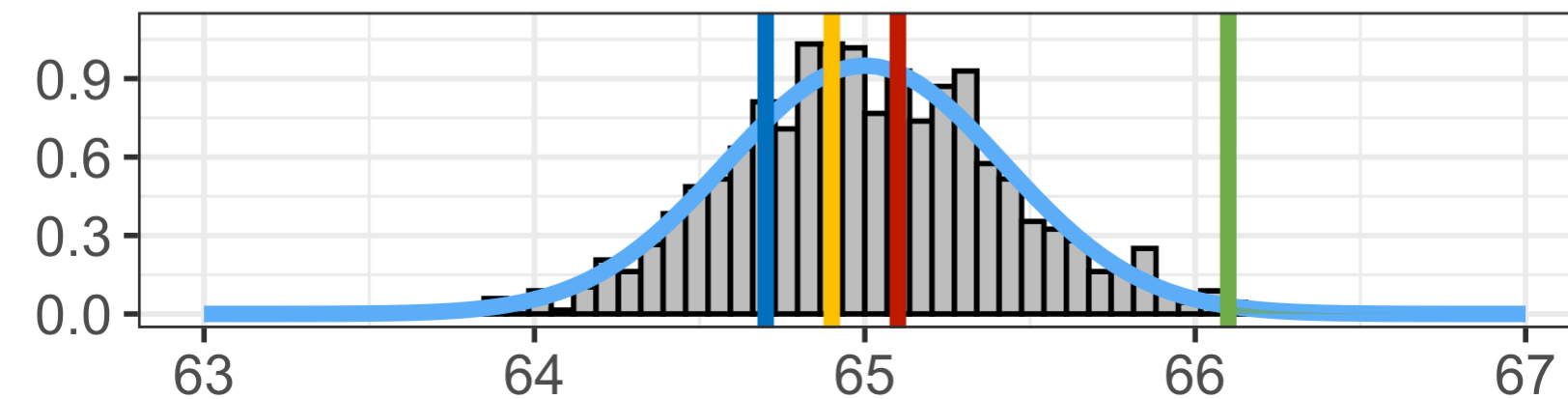
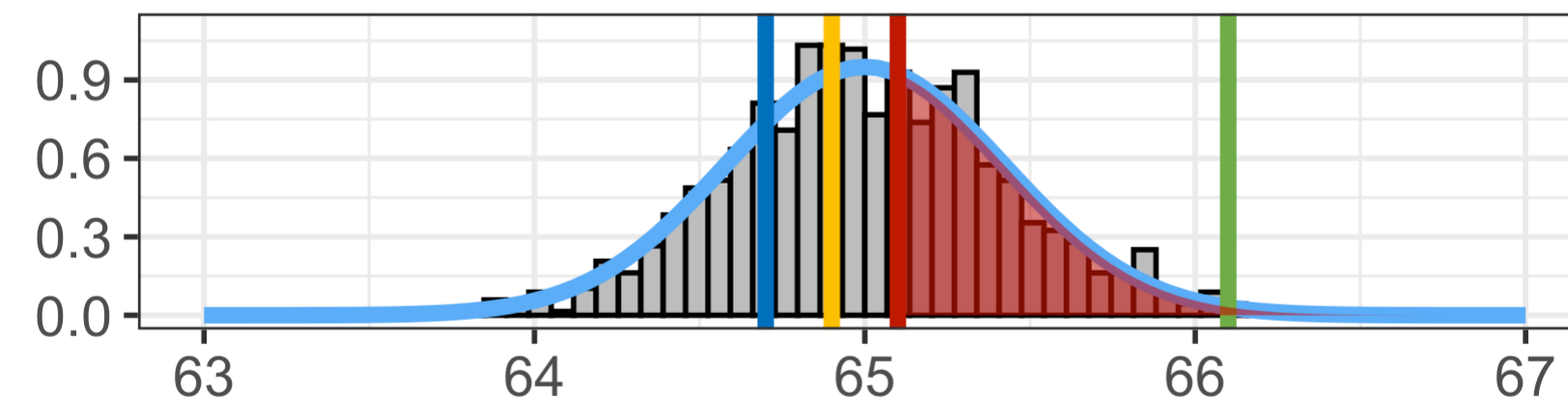
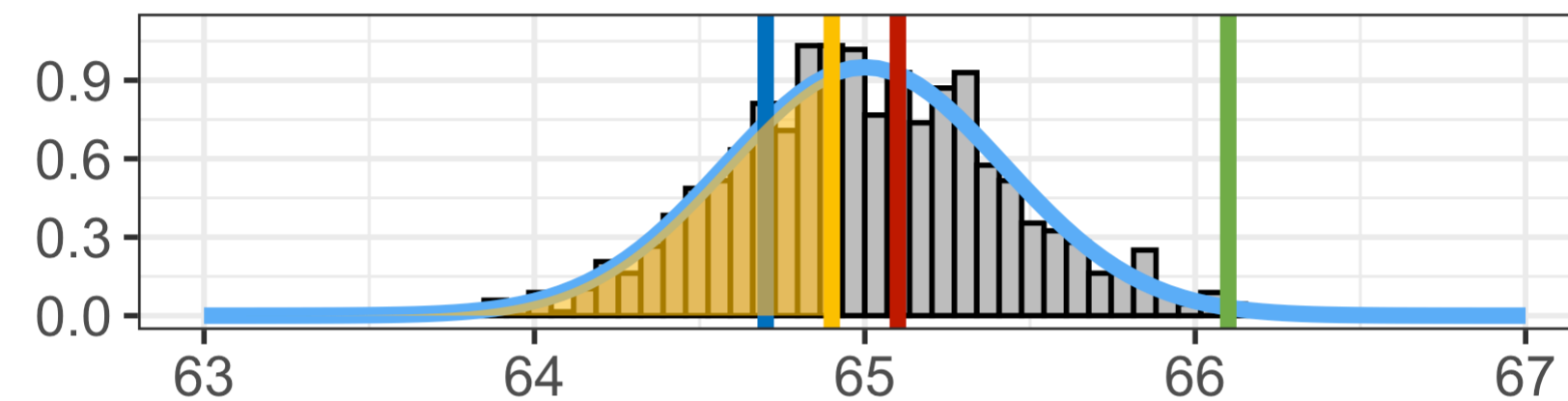
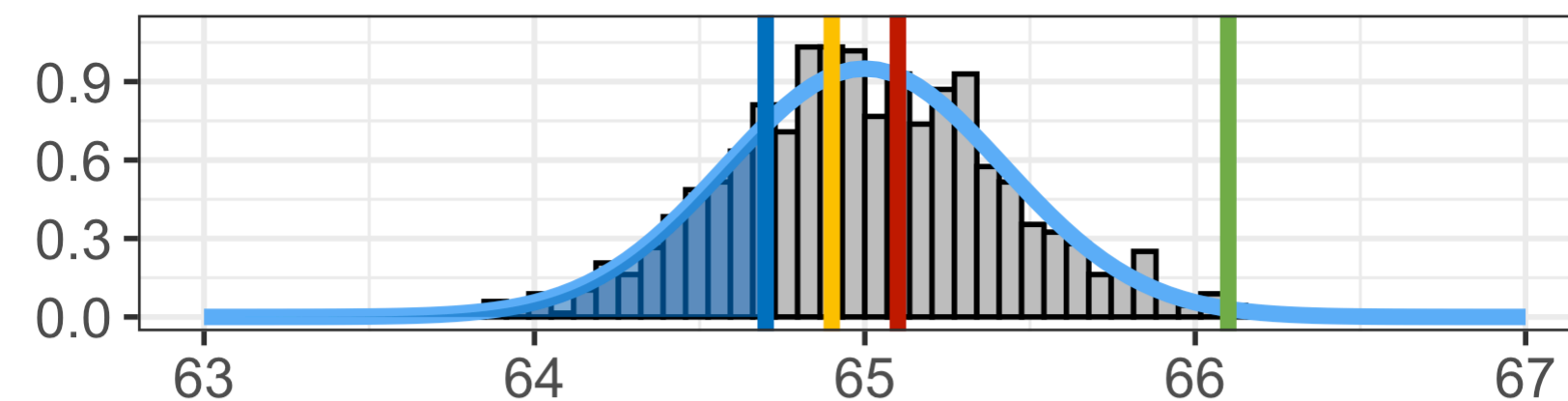
This time: Point estimates with probability assuming population mean μ

Sample 50 people
 $\bar{x} = 64.7, s = 3.1$

Sample 50 people
 $\bar{x} = 64.9, s = 3.2$

Sample 50 people
 $\bar{x} = 65.1, s = 2.8$

Sample 50 people
 $\bar{x} = 66.1, s = 3.4$



Assuming the population mean is μ (65 inches), what is the probability that we observe $\bar{x}$ or a more extreme sample mean?

Last time: Confidence interval (CI) for the mean μ (z vs. t)

- In summary, we have two cases that lead to different ways to calculate the confidence interval

Case 1: We know the population standard deviation

$$\bar{x} \pm z^* \times \text{SE}$$

- with $\text{SE} = \frac{\sigma}{\sqrt{n}}$ and σ is the population standard deviation
- For 95% CI, we use:
 - $z^* = \text{qnorm}(p = 0.975) = 1.96$

Case 2: We **do not** know the population sd

$$\bar{x} \pm t^* \times \text{SE}$$

- with $\text{SE} = \frac{s}{\sqrt{n}}$ and s is the sample standard deviation
- For 95% CI, we use:
 - $t^* = \text{qt}(p = 0.975, \text{df} = n-1)$

Poll Everywhere Question 1

This time: Hypothesis test (z vs. t)

- We have two different distributions from which we run a hypothesis test

Case 1: We know the population standard deviation

- We use a test statistic from a Normal distribution:

$$z_{\bar{x}} = \frac{\bar{x} - \mu}{SE}$$

- with $SE = \frac{\sigma}{\sqrt{n}}$ and σ is the population standard deviation

Case 2: We **do not** know the population sd

- We use a test statistic from a Student's t -distribution:

$$t_{\bar{x}} = \frac{\bar{x} - \mu}{SE}$$

- with $SE = \frac{s}{\sqrt{n}}$ and s is the sample standard deviation
- This is usually the case in real life

Is 98.6°F really the mean “healthy” body temperature?

- We will illustrate how to perform a hypothesis test as we work through this example
- Where did the 98.6°F value come from?
 - German physician Carl Reinhold August Wunderlich determined 98.6°F (or 37°C) based on temperatures from 25,000 patients in Leipzig in 1851.
- 1992 JAMA article by Mackowiak, Wasserman, & Levine
 - They claim that 98.2°F (36.8°C) is a more accurate average body temp
 - Sample: n = 148 healthy individuals aged 18 - 40 years
- Other research indicating that the human body temperature is lower
 - *Decreasing human body temperature in the United States since the Industrial Revolution*
 - *Defining Usual Oral Temperature Ranges in Outpatients Using an Unsupervised Learning Algorithm*
 - NYT article *The Average Human Body Temperature Is Not 98.6 Degrees*

Question: based on the 1992 JAMA data, is there evidence to support that the population mean body temperature is different from 98.6°F?

Question: based on the 1992 JAMA data, is there evidence to support that the population mean body temperature is different from 98.6°F?

Two approaches to answer this question:

Confidence interval

- Create a **confidence interval (CI)** for the population mean μ and determine whether 98.6°F is inside the CI or not.
- Answering the question: is 98.6°F a plausible value?

Hypothesis test

- Run a **hypothesis test** to see if there is evidence that the population mean μ is *significantly different* from 98.6°F or not
- This does not give us a range of plausible values for the population mean μ .
- Instead, we calculate a *test statistic* and *p-value*
- See how likely we are to observe the sample mean $\bar{x}$ or a more extreme sample mean assuming that the population mean μ is 98.6°F

Approach 1: Create a 95% CI for the population mean body temperature

- Use data based on the results from the 1992 JAMA study
 - The original dataset used in the JAMA article is not available
 - However, Allen Shoemaker from Calvin College created a **dataset** with the same summary statistics as in the JAMA article, which we will use:

$$\bar{x} = 98.25, s = 0.733, n = 130$$

CI for μ :

$$\begin{aligned} & \bar{x} \pm t^* \cdot \frac{s}{\sqrt{n}} \\ & 98.25 \pm 1.979 \cdot \frac{0.733}{\sqrt{130}} \\ & 98.25 \pm 0.127 \\ & (98.123, 98.377) \end{aligned}$$

$$\text{Used } t^* = \text{qt}(.975, \text{df}=129) = 1.979$$

Conclusion: We are 95% confident that the (population) mean body temperature is between 98.123°F and 98.377°F, which is discernably different than 98.6°F.

Approach 2: Hypothesis Test

From before:

- Run a **hypothesis test** to see if there is evidence that the population mean μ is *significantly different* from 98.6°F or not.
 - This does not give us a range of plausible values for the population mean μ .
 - Instead, we calculate a *test statistic* and *p-value*
 - to see how likely we are to observe the sample mean $\bar{x}$
 - or a more extreme sample mean
 - assuming that the population mean μ is 98.6°F.

How do we calculate a *test statistic* and *p-value*?

- Use the sampling distribution and central limit theorem!!
- Focus on Case 2: we don't know the population sd σ

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General steps in a Hypothesis Test

1. Check the **assumptions**
2. Set the **level of significance** α
3. Specify the **null** (H_0) and **alternative** (H_A) **hypotheses**
 1. In symbols
 2. In words
 3. Alternative: one- or two-sided?
4. Calculate the **test statistic**.
5. Calculate the **p-value** based on the observed test statistic and its sampling distribution
6. Write a **conclusion** to the hypothesis test
 1. Do we reject or fail to reject H_0 ?
 2. Write a conclusion in the context of the problem

Step 1: Check the assumptions

- The assumptions to run a hypothesis test on a sample are:
 - **Independent observations:** the observations were collected independently.
 - **Approximately normal sample or big n:** the distribution of the sample should be approximately normal, or the sample size should be at least 30
 - AKA: sample is approximately Normal OR we can use the CLT
- These are the criteria for the Central Limit Theorem in Lesson 09: Variability in estimates
- In our example, we would check the assumptions with a statement:
 - The individual observations are independent and the number of individuals in our sample is 130. Thus, we can use CLT to approximate the sampling distribution.

Step 2: Set the level of significance α

- **Before doing a hypothesis test**, we set a cut-off for how small the p -value should be in order to reject H_0 .
- It is important to specify how rare or unlikely an event must be in order to represent sufficient evidence against the null hypothesis.
- We call this the **significance level**, denoted by the Greek symbol **alpha (α)**
 - Typically choose $\alpha = 0.05$
- This is parallel to our confidence interval
 - α is the probability of rejecting the null hypothesis when it is true (it's a measure of potential error)
 - Null: our initial assumption about the mean (in example, $\mu = 98.6$)
 - From repeated $(1 - \alpha)\%$ confidence intervals, we will have about $\alpha\%$ intervals that do not cover μ even though they come from the distribution with mean μ

Step 3: Null & Alternative Hypotheses (1/2)

In statistics, a **hypothesis** is a statement about the value of an **unknown population parameter**.

A **hypothesis test** consists of a test between two competing hypotheses:

1. a **null** hypothesis H_0 (pronounced “H-naught”) vs.
2. an **alternative** hypothesis H_A (also denoted H_1)

Example of hypotheses in words:

H_0 : The population mean body temperature is 98.6 °F
vs. H_A : The population mean body temperature is not 98.6 °F

1. H_0 is a claim that there is “no effect” or “no difference of interest.”
2. H_A is the claim a researcher wants to establish or find evidence to support. It is viewed as a “challenger” hypothesis to the null hypothesis H_0

Step 3: Null & Alternative Hypotheses (2/2)

Notation for hypotheses

$$H_0 : \mu = \mu_0$$

vs. $H_A : \mu \neq, <, \text{or}, > \mu_0$

Hypotheses test for example

$$H_0 : \mu = 98.6$$

vs. $H_A : \mu \neq 98.6$

We call μ_0 the *null value* (hypothesized population mean from H_0)

$$H_A : \mu \neq \mu_0$$

- not choosing a priori whether we believe the population mean is greater or less than the null value μ_0

$$H_A : \mu < \mu_0$$

- believe the population mean is **less** than the null value μ_0

$$H_A : \mu > \mu_0$$

- believe the population mean is **greater** than the null value μ_0

- $H_A : \mu \neq \mu_0$ is the most common option, since it's the most conservative

Poll Everywhere Question 2

Step 4: Test statistic (& its distribution)

Case 1: We know the population standard deviation

- We use a test statistic from a Normal distribution:

$$z_{\bar{x}} = \frac{\bar{x} - \mu}{SE}$$

- with $SE = \frac{\sigma}{\sqrt{n}}$ and σ is the population standard deviation
- Statistical theory tells us that $z_{\bar{x}}$ follows a **Standard Normal distribution** $N(0, 1)$

Case 2: We do not know the population sd

- We use test statistic from Student's t-distribution:

$$t_{\bar{x}} = \frac{\bar{x} - \mu}{SE}$$

- with $SE = \frac{s}{\sqrt{n}}$ and s is the sample standard deviation
- Statistical theory tells us that $t_{\bar{x}}$ follows a **Student's t distribution** with degrees of freedom (df) = $n - 1$

$\bar{x}$ = sample mean, μ_0 = hypothesized population mean from H_0 ,
 σ = *population* standard deviation, s = *sample* standard deviation,
 n = sample size

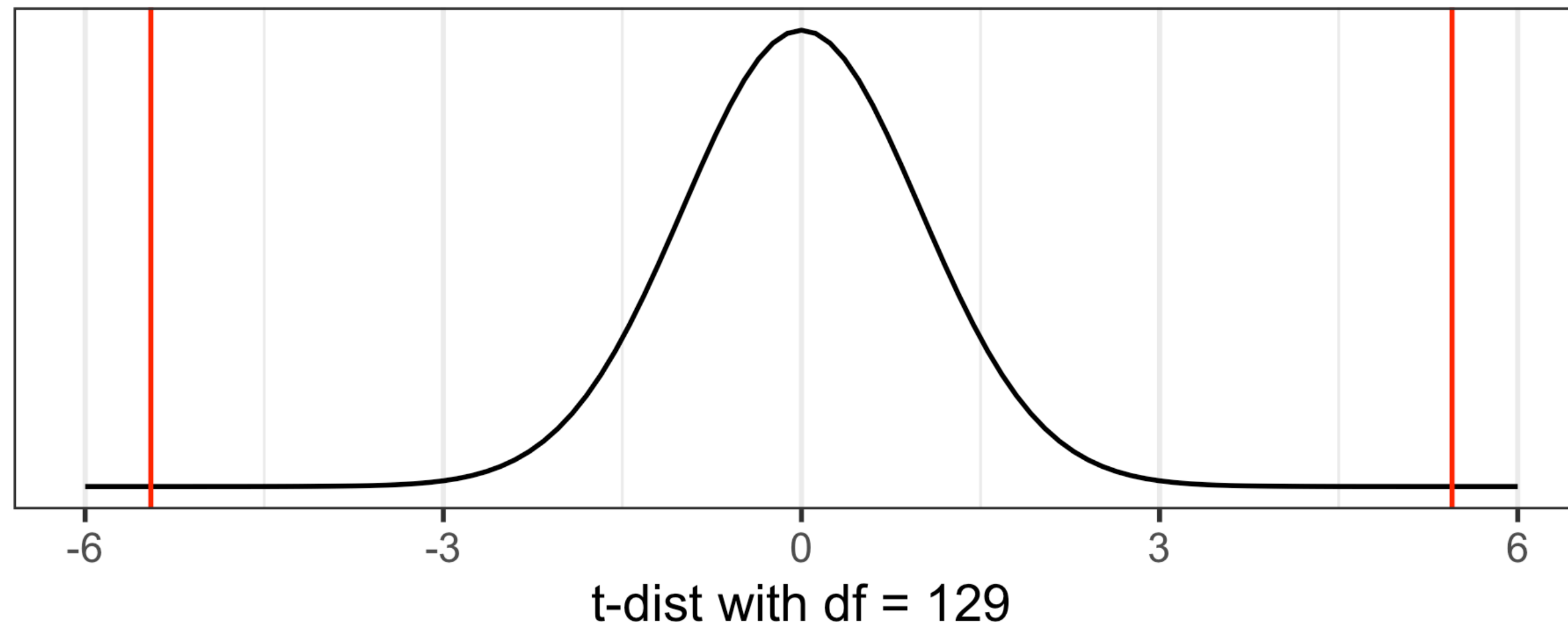
Step 4: Test statistic calculation

From our example: Recall that $\bar{x} = 98.25$, $s = 0.733$, and $n = 130$

The test statistic is:

$$t_{\bar{x}} = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{98.25 - 98.6}{\frac{0.73}{\sqrt{130}}} = -5.45$$

- Statistical theory tells us that $t_{\bar{x}}$ follows a **Student's t-distribution** with $df = n - 1 = 129$

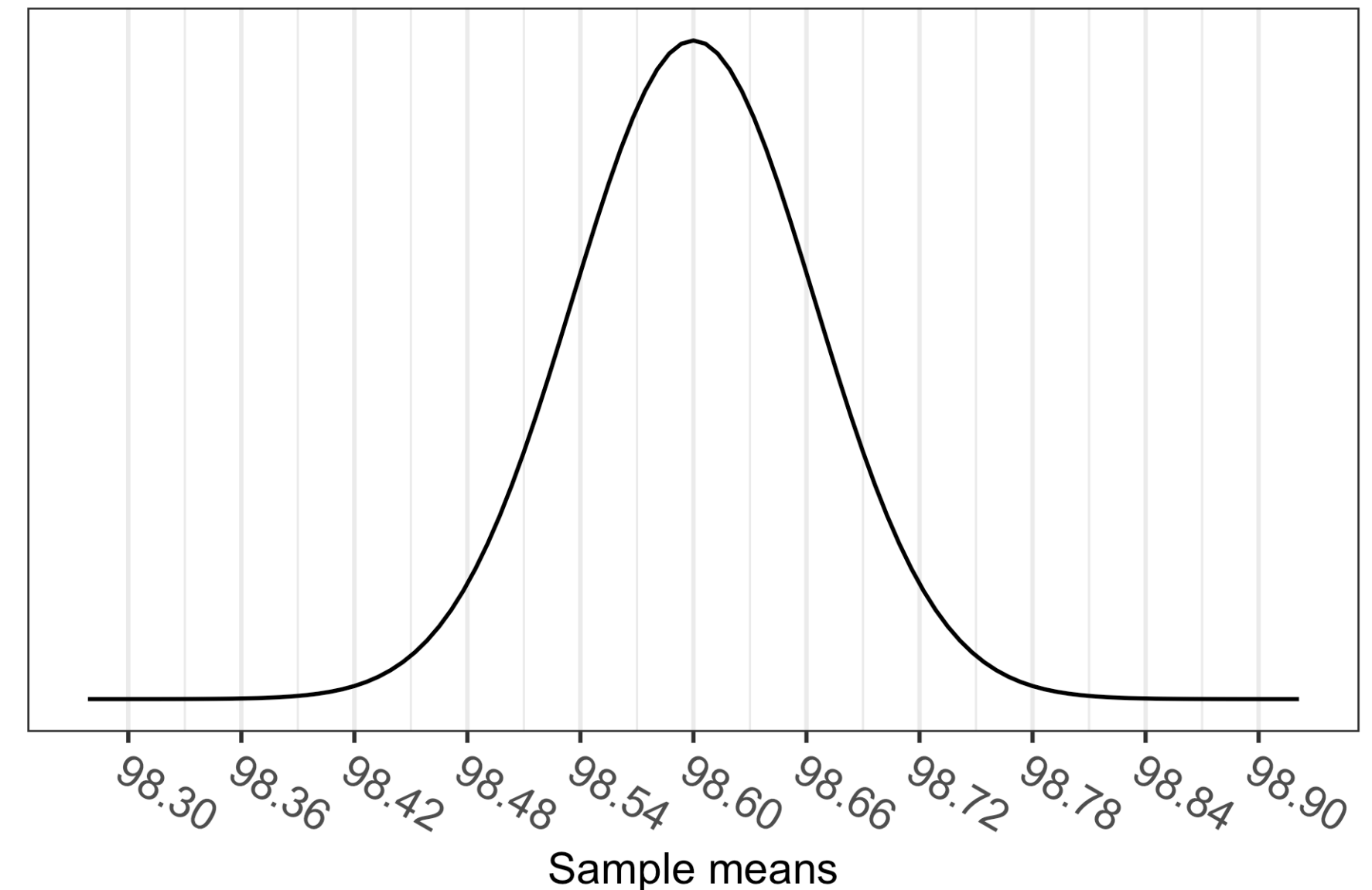


Step 5: p-value

The **p-value** is the **probability** of obtaining a test statistic *just as extreme or more extreme* than the observed test statistic assuming the null hypothesis H_0 is true.

- The p -value is a quantification of “surprise”
 - Assuming H_0 is true, *how surprised are we with the observed results?*
 - Ex: assuming that the true mean body temperature is 98.6°F, how surprised are we to get a sample mean of 98.25°F (or more extreme)?
- If the p -value is “small,” it means there’s a small probability that we would get the observed statistic (or more extreme) when H_0 is true.

Sampling distribution of mean body temperatures

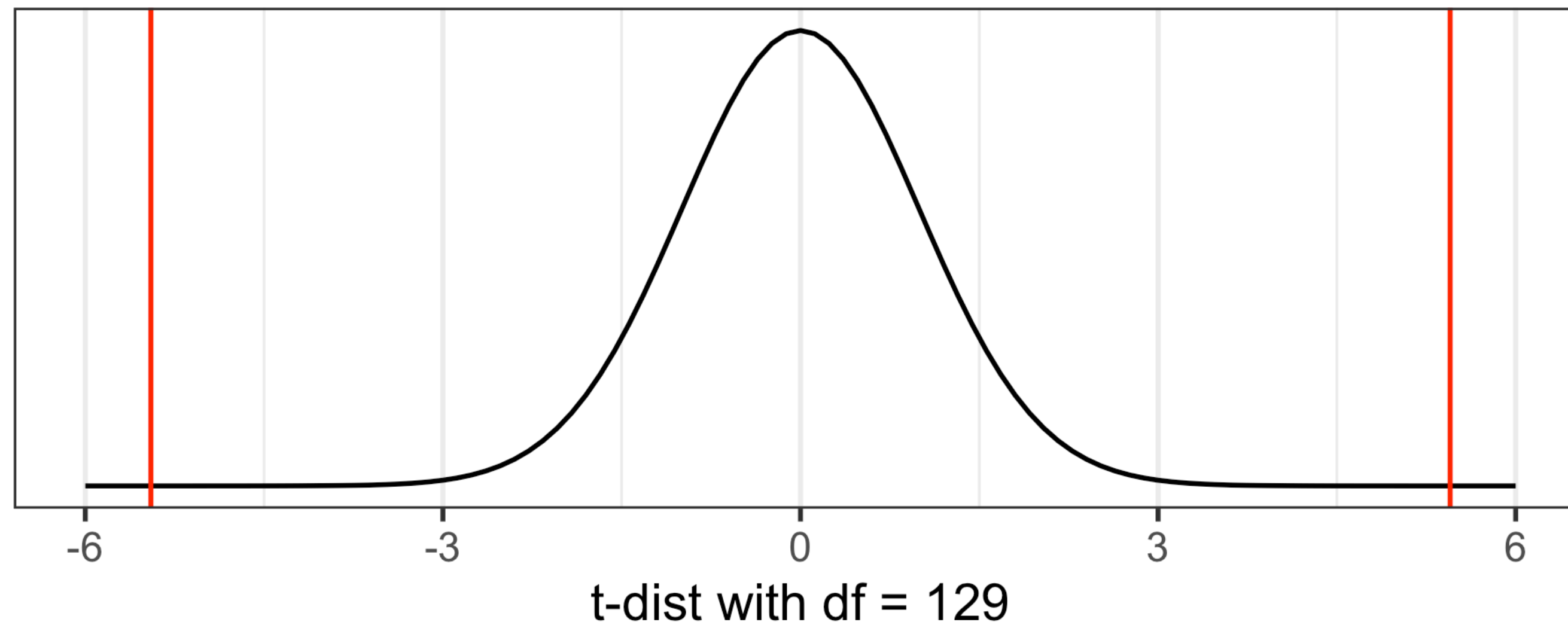


Step 5: p-value calculation

Calculate the p -value using the **Student's t-distribution** with $df = n - 1 = 130 - 1 = 129$:

$$p\text{-value} = P(T \leq -5.45) + P(T \geq 5.45) = 2.410889 \times 10^{-07}$$

```
1 # use pt() instead of pnorm()  
2 # need to specify df  
3 2*pt(-5.4548, df = 130-1, lower.tail = TRUE)  
[1] 2.410889e-07
```



Step 6: Conclusion to hypothesis test

$$H_0 : \mu = \mu_0$$

vs. $H_A : \mu \neq \mu_0$

- Need to compare p-value to our selected $\alpha = 0.05$
- Do we reject or fail to reject H_0 ?

If p-value $< \alpha$, reject the null hypothesis

- There is sufficient evidence that the (population) mean body temperature is discernibly different from μ_0 (p-value = ____)
- The average (insert measure) in the sample was $\bar{x}$ (95% CI ,), which is discernibly different from μ_0 (p-value = ____).

If p-value $\geq \alpha$, fail to reject the null hypothesis

- There is insufficient evidence that the (population) mean body temperature is discernibly different from μ_0 (p-value = ____)
- The average (insert measure) in the sample was $\bar{x}$ (95% CI ,), which is not discernibly different from μ_0 (p-value = ____).

Step 6: Conclusion to hypothesis test

$$H_0 : \mu = 98.6$$

vs. $H_A : \mu \neq 98.6$

- Recall the p -value = 2.410889×10^{-07}
- Need to compare back to our selected $\alpha = 0.05$
- Do we reject or fail to reject H_0 ?

Conclusion statement:

- Basic: (“stats class” conclusion)
 - There is sufficient evidence that the (population) mean body temperature is discernibly different from 98.6°F (p -value < 0.001).
- Better: (“manuscript style” conclusion)
 - The average body temperature in the sample was 98.25°F (95% CI 98.12, 98.38°F), which is discernibly different from 98.6°F (p -value < 0.001).

Poll Everywhere Question 3

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2. Determine if a single-sample mean is different than a population mean using a hypothesis test.
3. Use R to calculate the test statistic, p-value, and confidence interval for a single-sample mean.

Load the dataset

- The data are in a csv file called `BodyTemperatures.csv`

```
1 library(here)    # first install this package
2
3 BodyTemps <- read.csv(here::here("data", "BodyTemperatures.csv"))
4 #               location: look in "data" folder
5 #               for the file "BodyTemperatures.csv"
6
7 glimpse(BodyTemps)
```

Rows: 130

Columns: 3

\$ Temperature <dbl> 96.3, 96.7, 96.9, 97.0, 97.1, 97.1, 97.1, 97.2, 97.3, 97.4...

\$ Gender <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...

\$ HeartRate <int> 70, 71, 74, 80, 73, 75, 82, 64, 69, 70, 68, 72, 78, 70, 75...

t.test: base R's function for testing one mean

- Use the body temperature example with $H_A : \mu \neq 98.6$
- We called the dataset `BodyTemps` when we loaded it

```
1 (temps_ttest <- t.test(x = BodyTemps$Temperature,  
2       alternative = "two.sided", # default setting  
3       mu = 98.6))
```

One Sample t-test

```
data: BodyTemps$Temperature  
t = -5.4548, df = 129, p-value = 2.411e-07  
alternative hypothesis: true mean is not equal to 98.6  
95 percent confidence interval:  
 98.12200 98.37646  
sample estimates:  
mean of x  
 98.24923
```

Note that the test output also gives the 95% CI using the t-distribution.

tidy() the t.test output

- Use the `tidy()` function from the `broom` package for briefer output in table format that's stored as a `tibble`
- Combined with the `gt()` function from the `gt` package, we get a nice table

```
1 tidy(temps_ttest) %>%  
2   gt() %>%  
3   tab_options(table.font.size = 40) # use a different size in your HW
```

estimate	statistic	p.value	parameter	conf.low	conf.high	method	alternative
98.24923	-5.454823	2.410632e-07	129	98.122	98.37646	One Sample t-test	two.sided

- Since the `tidy()` output is a `tibble`, we can easily `pull()` specific values from it:

Using base R's `$`

```
1 tidy(temps_ttest)$p.value  
[1] 2.410632e-07
```

What's next?

CI's and hypothesis testing for different scenarios:

Lesson	Section	Population parameter	Symbol (pop)	Point estimate	Symbol (sample)
12	5.1	Pop mean	μ	Sample mean	$\bar{x}$
13	5.2	Pop mean of paired diff	μ_d or δ	Sample mean of paired diff	$\bar{x}_d$
14	5.3	Diff in pop means	$\mu_1 - \mu_2$	Diff in sample means	$\bar{x}_1 - \bar{x}_2$
16	8.1	Pop proportion	p	Sample prop	$\hat{p}$
16	8.2	Diff in pop prop's	$p_1 - p_2$	Diff in sample prop's	$\hat{p}_1 - \hat{p}_2$

Reference: what does it all look like together?

Example of hypothesis test based on the 1992 JAMA data

Is there evidence to support that the population mean body temperature is different from 98.6°F?

1. **Assumptions:** The individual observations are independent and the number of individuals in our sample is 130. Thus, we can use CLT to approximate the sampling distribution.

2. Set $\alpha = 0.05$

3. **Hypothesis:**
 $H_0 : \mu = 98.6$
vs. $H_A : \mu \neq 98.6$

4-5.

```
1 temps_ttest <- t.test(x = BodyTemps$Temperature, mu = 98.6)
2 tidy(temps_ttest) %>% gt() %>% tab_options(table.font.size = 36)
```

estimate	statistic	p.value	parameter	conf.low	conf.high	method	alternative
98.24923	-5.454823	2.410632e-07	129	98.122	98.37646	One Sample t-test	two.sided

6. **Conclusion:** We reject the null hypothesis. The average body temperature in the sample was 98.25°F (95% CI 98.12, 98.38°F), which is discernibly different from 98.6°F (p -value < 0.001).